Drug Repurposing Hypothesis Assessment: MALT1 Protease Inhibitor (Safimaltib) in Sjögren's Syndrome

Summary

The case is mechanistically plausible but weak at the drug level, narrowly applicable, and burdened by a population-specific safety liability. The pathway the drug targets — CBM-complex–driven, A20/NF-κB–regulated B-cell signalling — is genuinely active in Sjögren's syndrome (SjS). Impaired A20 (TNFAIP3) function is genetically and functionally enriched in SjS-associated lymphoma, and epithelial-specific A20 loss reproduces Sjögren-like glandular disease in mice. These are real, human-validated anchors for the rationale. However, the mechanism is concentrated in a biomarker-identifiable minority of patients — those with ectopic germinal-center-like structures, cryoglobulinemia, and other lymphoma-risk features — not in the broad SjS population that trials typically enrol.

Two pillars of the specific hypothesis do not survive scrutiny. First, one premise is factually wrong for SjS: Sjögren-associated MALT lymphomas largely **lack** the classic MALT1 translocations (e.g., t(11;18)/API2-MALT1) that the hypothesis invokes as a driver; whole-exome and FISH studies show a low mutational burden and absence of the recurrent MALT-lymphoma aberrations. Second, no MALT1 protease inhibitor has ever been tested in Sjögren's; safimaltib (JNJ-67856633) is validated only in relapsed/refractory B-cell malignancy, where its first-in-human Phase 1 trial produced a modest objective response rate (28.8%) and a toxicity profile dominated by hepatic and hematologic — not overt autoimmune — adverse events over the trial window.

The closest clinical proof-of-concept for the upstream pathway, the BTK inhibitor remibrutinib, met an objective disease-activity endpoint (ΔESSDAI −2.86, p=0.003) but produced **no** symptom benefit, and B-cell depletion with rituximab similarly modified gland biology while failing patient-reported endpoints in unselected patients. The decisive counter-argument is safety: MALT1 protease inhibition reproducibly depletes regulatory T cells and provokes IPEX-like autoimmunity in animals — a paradoxical liability when the proposed indication is itself an

autoimmune disease. On balance, the pathway is real, the drug's benefit in SjS is an untested extrapolation from DLBCL, and the risk/benefit favours a narrow, biomarker-enriched experimental test rather than a broad repurposing programme.

Overall verdict: Plausible mechanism, weak drug-level case, significant safety counter-argument — a conditional "worth a narrowly targeted test," not a broad repurposing opportunity.

Key Findings

Finding 1 — Impaired A20/NF-κB control is genetically linked to Sjögren's-associated lymphoma (established)

A germline variant of **TNFAIP3** (the gene encoding the NF-κB brake A20), rs2230926G, is associated with primary SjS (pSS)–associated lymphoma. A meta-analysis of UK and French cohorts found an odds ratio of **2.48 (95% CI 1.87–3.28, p=0.0037)** comparing pSS patients with lymphoma to those without. Functional A20 abnormalities (germline or somatic) were present in **60%** of pSS-lymphoma patients, rising to **77%** among those with mucosa-associated lymphoid tissue (MALT) lymphoma — precisely the drug's intended target population.

- PMID: 26338037: *"The meta-analysis of the two cohorts confirmed these results (OR=2.48, 95% CI (1.87 to 3.28) p=0.0037."*
- PMID: 24159176: *"The frequency was even higher (77%) among pSS patients with mucosa-associated lymphoid tissue lymphoma."*

Interpretation (established + inference): Because A20 is the principal negative regulator of NF-κB, its impairment places the pathway the drug targets causally upstream of pSS lymphomagenesis. This is the single strongest human-genetics anchor for the hypothesis.

Finding 2 — Epithelial A20 loss reproduces Sjögren-like disease, validating NF-κB as a driver (established)

Keratin14^{+} epithelial-specific A20 knockout mice developed T-cell–dominated leukocytic infiltration, immune foci, lymphoepithelial lesions, elevated IFNγ/TNFα/IL-6/CXCL10/CXCL13, and functionally reduced saliva volume and mucin content — recapitulating the early clinical picture of SjS.

- PMID: 30067782: *"We show that immune activated epithelial cells can cause T cell dominated leukocytic infiltration and immune foci development of the SGs, reflecting the early clinical picture."*

Interpretation (established + inference): De-repression of NF-κB is sufficient to drive Sjögren-like gland pathology. Importantly, this model locates a substantial part of the disease-initiating NF-κB signal in the **epithelial** compartment, not the B cell — a nuance that bears directly on whether a B/T-cell–focused MALT1 inhibitor hits the right cells (see Finding 7 and Limitations).

Finding 3 — MALT1 protease inhibition causes Treg loss and IPEX-like autoimmunity: a population-specific safety liability (established)

In rat and dog, the selective MALT1 inhibitor MLT-943 caused rapid, dose-dependent Treg reduction and progressive IPEX-like pathology (severe intestinal inflammation, high serum IgE, systemic T-cell activation) despite efficacy in collagen-induced arthritis. Germline MALT1 protease-dead mice develop lethal multi-organ inflammation. A 2025 oncology review notes that safimaltib itself disrupts immunosuppressive Treg function. A mitigating nuance exists: adult-onset (rather than germline) MALT1 inhibition is better tolerated, suggesting the hazard is dose/schedule-dependent rather than absolute.

- PMID: 32425939: *"pharmacological inhibition of MALT1 was associated with a rapid and dose-dependent reduction in Tregs and resulted in the progressive appearance of immune abnormalities and clinical signs of an IPEX-like pathology."*
- PMID: 41783325: *"it also disrupts immunosuppressive Treg function, risking autoimmune toxicity by compromising the tumor microenvironment."*

Interpretation (established + inference): This is the hypothesis's sharpest counter-argument. A drug that induces autoimmunity by depleting Tregs is a paradoxical choice for an autoimmune disease. The risk is directly tied to the drug's on-target mechanism and is therefore difficult to engineer away, though adult-onset dosing may blunt it.

Finding 4 — No MALT1-inhibitor trial exists in SjS; the closest upstream blockade gives modest objective but no symptom benefit (established)

Safimaltib (JNJ-67856633) is in early-phase trials only for relapsed/refractory B-cell malignancies, not SjS. The BTK inhibitor remibrutinib — upstream of the CBM/MALT1 node — met its primary endpoint in a Phase 2 pSS RCT (NCT04035668): **ΔESSDAI −2.86, p=0.003**, but produced **no ESSPRI (symptom) benefit** (Δ0.17, p=0.663). A network meta-analysis (27 RCTs, n=2261) shows pathway-adjacent agents give only modest ESSDAI reductions.

- PMID: 37932009: *"Remibrutinib significantly improved ESSDAI score in patients with SjS over 24 weeks compared with placebo (ΔESSDAI -2.86, p=0.003). No treatment effect was observed in ESSPRI score."*
- PMID: 42113283: *"remibrutinib (MD = -2.86, 95% CrI:-4.92, -0.8)."*

Interpretation (established + inference): The best available clinical read-through for the BCR→NF-κB axis shows the effect is modest and confined to objective disease-activity measures, not patient-reported symptoms. This is a cautionary signal for what a MALT1 inhibitor — acting one node downstream — could realistically achieve, and it highlights the endpoint problem.

Agent	Target (relative to CBM/MALT1 node)	ESSDAI effect	Symptom (ESSPRI) effect
Remibrutinib	BTK (upstream)	−2.86 (p=0.003)	None (Δ0.17, p=0.663)
Iscalimab	CD40 (upstream/costim)	−3.4	—
Ianalumab	BAFF-R + B-cell depletion	−1.85	—
Rituximab (TRACTISS)	CD20 B-cell depletion	Gland biology modified	Co-primary symptom endpoints not met

Finding 5 — The lymphoma-prone NF-κB mechanism is confined to a biomarker-identifiable subgroup (established)

Ectopic germinal-center (GC)-like structures in minor salivary gland biopsy confer a **7.8-fold increased lymphoma risk (95% CI 1.73–34.86, p=0.0075)**. Other independent predictors include cryoglobulinemia (HR 7.1), male sex, sensorimotor neuropathy, and splenomegaly. In a Swedish cohort, 6 of 7 patients who developed non-Hodgkin lymphoma were GC-positive at diagnosis (p=0.001). Only ~16–25% of pSS patients carry GC-like structures.

- PMID: 29669392: *"The presence of ectopic GC-like structures was associated with a 7.8-fold increased risk of lymphoma occurrence (95% confidence interval [95% CI] 1.73-34.86 [P = 0.0075])."*
- PMID: 21715359: *"Six of the seven patients had GC-like structures at diagnosis."*

Interpretation (established + inference): Any realistic MALT1-inhibitor trial in SjS must enrich for this high-risk endotype. A broad, unselected trial would dilute whatever mechanism-specific effect exists and is likely to fail.

Finding 6 — SjS is biologically heterogeneous; MALT1 inhibition maps to one axis, not the whole disease (established)

A 2026 framework organizes SjS into three axes — an interferon-driven systemic axis, a **B-cell/lymphoproliferative** axis, and a symptom/fibro-structural axis — each with distinct biomarkers, histopathology, endpoints, and therapeutic targets. Recent positive trials became detectable only once stratification was aligned to biology.

- PMID: 42449965: *"a B-cell/lymphoproliferative axis, and a symptom/fibro-structural axis. Each axis carries its own characteristic biomarkers, histopathology, prognostic features, candidate endpoints, and therapeutic targets."*
- PMID: 42449965: *"meaningful clinical benefit becomes detectable once stratification is aligned to biology."*

Interpretation (established + inference): MALT1/NF-κB targeting maps to the B-cell/lymphoproliferative axis only. The symptom axis (dryness, fatigue) — which dominates patient-reported endpoints — is unlikely to respond, explaining why upstream agents (Finding 4) improve ESSDAI but not ESSPRI.

Finding 7 — MALT1 paracaspase activity is validated mainly in ABC-DLBCL; transfer to SjS is plausible but unproven (established)

MALT1 inhibitors suppress ABC-DLBCL growth in vitro and in vivo, where these lymphomas are biologically dependent on CBM-driven NF-κB. No study has tested MALT1 inhibition in Sjögren-associated MALT lymphoma or in non-malignant pSS B cells. The original molecular-pathways authors frame autoimmune use as a prospective "perhaps."

- PMID: 24004675: *"Newly developed MALT1 small-molecule inhibitors suppress the growth of ABC-DLBCLs in vitro and in vivo."*
- PMID: 24004675: *"deploying MALT1 inhibitors for the treatment of B-cell lymphomas and perhaps autoimmune diseases that involve increased B- or T-cell receptor signaling."*

Interpretation (established + inference): The entire efficacy base sits in aggressive lymphoma biology. ABC-DLBCL (chronically-active-BCR, aggressive) differs from indolent antigen-driven salivary MALT lymphoma and from non-malignant gland inflammation. Extrapolating to a chronic, indolent autoimmune exocrinopathy is mechanistically reasonable (shared CBM→MALT1 node) but has never been demonstrated in Sjögren's tissue.

Finding 8 — B-cell depletion modifies gland biology but failed symptomatic endpoints in unselected SjS (established)

In TRACTISS (ISRCTN65360827, n=110), rituximab prevented worsening of salivary-gland inflammation by inhibiting class-switched memory B-cell accumulation and down-regulated immunoglobulins and lymphoid-organization genes; an ultrasound substudy showed significant improvement (between-group difference −1.2, p=0.0099 at wk16; p=0.0023 at wk48). STAR-responders had higher baseline CXCL13, IL-22, IL-17A/F and TNF-α. However, the co-primary patient-reported endpoints (fatigue, oral dryness) were not met.

- PMID: 38073013: *"Rituximab treatment prevented the worsening of SG inflammation observed in the placebo arm, by inhibiting the accumulation of class-switched memory B cells within the SG."*
- PMID: 38073013: *"patients classified as responders according to STAR displayed significantly higher baseline levels of C-X-C motif chemokine ligand-13 (CXCL13), interleukin (IL)-22, IL-17A, IL-17F, and tumor necrosis factor-α (TNF-α)."*

Interpretation (established + inference): The B-cell compartment in the gland is pharmacologically reachable and its biology is modifiable, but modifying it does not reliably relieve symptoms. Baseline inflammatory biomarkers stratify responders — reinforcing the enrichment strategy a MALT1 trial would need.

Finding 9 — Sjögren-associated MALT lymphomas largely lack the classic MALT1 translocations: the premise's "MALT1 translocation" driver is wrong for pSS (established — refutes part of the hypothesis)

Whole-exome sequencing of 17 pSS-associated parotid MALT lymphomas found a low mutational burden (median 7 variants/case, range 2–78) and that the recurrent genomic aberrations of MALT lymphoma were not associated with pSS. Earlier FISH series of salivary MALT lymphoma found only 3/28 with t(11;18) and 2/28 trisomy 18, with no IgH or BCL10 translocations.

- PMID: 35205758: *"Unlike recurrent genomic aberrations observed in MALT lymphoma, which were not associated with pSS (non-pSS), it is unknown which somatic aberrations underlie the development of pSS-associated MALT lymphomas."*
- PMID: 19961773: *"Interphase FISH showed that 3 cases harbored t(11;18) and 2 cases harbored trisomy 18, but none of all found IgH and bcl-10 translocations."*

Interpretation (established + inference): The hypothesis explicitly names "MALT1 translocations" as a disease driver. For SjS this is largely incorrect: the translocations that constitutively activate MALT1 in sporadic MALT lymphoma are rare in Sjögren-associated disease. The pSS mechanism is better explained by A20/NF-κB dysregulation and chronic BCR stimulation. Notably, translocation-negative tumours remain antigen/BCR-driven, which *keeps* them dependent on the upstream CBM→MALT1 node — so this refutes the stated rationale yet does not fully close the mechanistic door.

Finding 10 — Safimaltib Phase 1 (NCT03900598): modest efficacy, hematologic/hepatic (not overt autoimmune) toxicity, and BCR-mutation responders (established)

The first-in-human dose-escalation/expansion study enrolled 226 R/R B-NHL/CLL patients; the recommended Phase 2 dose was 300 mg once-daily oral. The most common adverse events were hyperbilirubinemia (48.2%, off-target), anemia (36.7%), neutropenia (31.4%), and

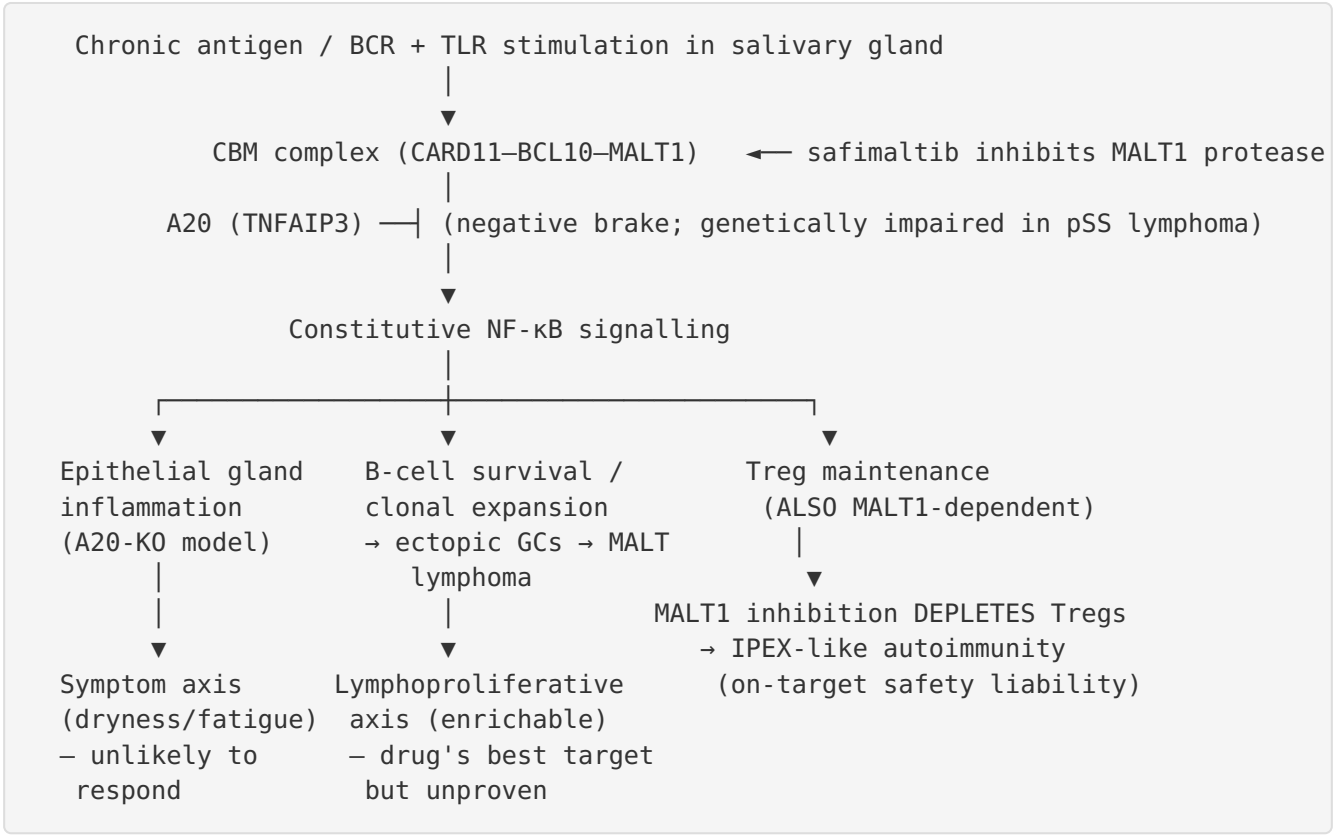
thrombocytopenia (27.4%); dose reduction mitigated AEs. Objective response rate was **28.8% (95% CI 23.0–35.1)**, with 3.9% complete responses. Notably, all 5 non-GCB DLBCL patients carrying ≥1 BCR-pathway-activating mutation responded.

- PMID: 42814116: *"Most common adverse events (AEs) were hyperbilirubinemia (48.2%; off-target related), anemia (36.7%), neutropenia (31.4%), and thrombocytopenia (27.4%)."*
- PMID: 42814116: *"All patients with non-germinal-center B-cell-like DLBCL and ≥1 B-cell-receptor pathway activating mutation (n=5) responded to safimaltib."*
- PMID: 42814116: *"Objective response rate was 28.8% (95% CI: 23.0%-35.1%)."*

Interpretation (established + inference): Two signals matter for repurposing. First, over the Phase 1 window the dominant toxicities were hepatic/hematologic rather than overt autoimmunity — reassuring on short-term tolerability, but silent on the chronic Treg-depletion risk that would matter in a years-long autoimmune indication. Second, the striking responder enrichment among BCR-pathway–mutant tumours validates a biomarker-driven selection strategy and is directly transferable to how a SjS trial should be designed.

Mechanistic Model / Interpretation

The pieces assemble into a coherent, but narrow, picture:



Three structural problems emerge from this model:

1. **Right pathway, partly wrong cell.** The A20-knockout data place a large share of the disease-initiating NF-κB signal in the gland **epithelium** (Finding 2), while a MALT1 inhibitor principally acts on lymphocyte antigen-receptor signalling. The drug may blunt the B-cell/lymphoproliferative arm while leaving the epithelial inflammatory arm — and the symptoms it drives — intact. Epithelial NF-κB is also driven by TLR/EDAR/TNFR inputs that do not all route through MALT1 (redundancy).
2. **Right pathway, wrong premise detail.** The hypothesis invokes MALT1 translocations as a driver, but these are rare in pSS-associated MALT lymphoma (Finding 9). The drug could still be rational as a block on chronic BCR→NF-κB signalling, but not for the reason stated.
3. **On-target benefit and on-target harm share the same node.** MALT1 sustains both pathogenic B-cell signalling and protective Treg function (Findings 3, 10). Inhibiting it to treat autoimmunity risks worsening autoimmunity — a mechanistic double-bind that cannot be fully separated by dosing, though adult-onset schedules may help.

The clinical read-throughs (Findings 4, 8) tie the model to reality: upstream (BTK) and parallel (CD20) interventions modify objective gland biology and disease-activity scores but do not relieve patient-reported symptoms, and only in enriched/unselected subgroups respectively. A MALT1 inhibitor would inherit this endpoint problem.

Evidence Base

PMID	Role in assessment	Direction
26338037	Germline TNFAIP3 variant → pSS lymphoma (OR 2.48)	**Supports** mechanism
24159176	A20 abnormalities in 77% of pSS-MALT lymphoma	**Supports** mechanism
30067782	Epithelial A20 loss → Sjögren-like disease	**Supports** (but epithelial locus)
32425939	MALT1 inhibition → Treg loss, IPEX-like pathology	**Against** (safety)
41783325	Safimaltib disrupts Treg function; validated in oncology only	**Against** / context
37932009	Remibrutinib: ESSDAI yes, ESSPRI no (NCT04035668)	**Mixed** (proof-of-concept + endpoint warning)
42113283	Network meta-analysis: modest ESSDAI effects	Context
29669392	Ectopic GCs → 7.8× lymphoma risk (enrichable subgroup)	**Supports** stratification
21715359	6/7 NHL patients GC-positive at diagnosis	**Supports** stratification
42449965	Three-axis SjS framework; stratify to detect benefit	Context / design
24004675	MALT1 inhibitors validated in ABC-DLBCL; autoimmune use "perhaps"	**Against** (unproven transfer)
38073013	TRACTISS: gland biology modified, symptoms not; biomarker responders	**Mixed**
35205758	pSS-MALT lymphomas lack classic MALT1 aberrations	**Refutes** premise detail
19961773	Salivary MALT lymphoma: t(11;18) rare, no BCL10/IgH	**Refutes** premise detail
42814116	Safimaltib Phase 1: ORR 28.8%, hepatic/heme AEs, BCR-mutant responders (NCT03900598)	**Mixed** (modest efficacy + enrichment signal)

Limitations and Knowledge Gaps

- **No direct data in SjS.** There is zero clinical or tissue-level experimental evidence of MALT1 inhibition in Sjögren's — malignant or non-malignant. Every efficacy inference is extrapolated from DLBCL (P 24004675). P 42814116
- **Cell-compartment uncertainty.** The relative contribution of epithelial vs. lymphocyte NF-κB to clinically meaningful SjS features is unresolved. If epithelial NF-κB dominates symptom biology (P 30067782), a lymphocyte-acting MALT1 inhibitor may underperform even where it engages its target.
- **Chronic-dosing safety is unknown.** The Phase 1 window (P 42814116) is too short to surface the Treg-depletion/autoimmunity risk predicted by preclinical models (P 32425939). This is the gap most likely to sink an autoimmune indication.
- **Endpoint mismatch.** The dominant SjS trial endpoints (ESSPRI symptoms) failed to move even with upstream BTK blockade (P 37932009) and B-cell depletion (P 38073013). A trial could show true biological effect yet "fail."
- **Premise partially falsified.** The specific "MALT1 translocation" driver named in the hypothesis is not operative in most pSS lymphoma (P 35205758; P 19961773) the rationale must be restated as chronic BCR→NF-κB dependence to remain defensible.

- **Reliance on review/secondary sources.** Some drug-level context (P 41783325) is from a review rather than primary trial data; the Treg-autoimmunity liability is strongest for MLT-943 and genetic models, which are not safimaltib itself. ClinicalTrials.gov was not queried directly; the absence of a SjS registration is inferred from the literature search.

Proposed Follow-up Experiments / Actions

1. **Ex vivo mechanistic test (highest value, lowest cost).** Treat minor-salivary-gland biopsy explants and sorted gland-resident B cells from GC-positive, cryoglobulinemic/low-C4 pSS patients with safimaltib; read out (i) MALT1 substrate cleavage and NF-κB target-gene suppression in gland B cells, (ii) reduction of CXCL13/BAFF/lymphoid-organization transcripts, versus (iii) the effect on local and circulating Treg frequency/function. This directly tests whether the drug blunts pathogenic B-cell NF-κB without disproportionately harming Tregs — the central risk/benefit question — before any human exposure.
2. **Biomarker-enriched Phase 1b/2a, not a broad trial.** Enrol only the lymphoproliferative endotype (ectopic GC-like structures on biopsy, cryoglobulinemia, low C4, elevated CXCL13/β2-microglobulin; (P 29669392). (P 38073013) Mirror the oncology responder signature — BCR-pathway activation (P 42814116). Primary endpoints should be **biological/objective** (salivary gland ultrasound, histologic focus score, B-cell and NF-κB cytokine panels, ESSDAI biological domains) rather than ESSPRI, which has repeatedly failed to detect pathway effects.
3. **Mandatory Treg pharmacodynamic monitoring.** Any human study must prospectively track circulating and tissue Treg number/function and IgE as early safety signals, given (P 32425939)

and 41783325. Pre-specify stopping rules. Favour adult-onset, intermittent dosing given 's tolerability signal.

P 33083726

4. **Comparative positioning against a safer upstream option.** Given remibrutinib's cleaner autoimmune-safety logic and its (modest) ESSDAI signal ( ), the key decision is whether moving one node downstream to MALT1 adds efficacy worth its Treg liability. A head-to-head pharmacodynamic comparison in the same enriched population would answer this most efficiently.

P 37932009

Verdict

The case is weak-to-moderate and should not advance as a broad repurposing effort. The target pathway is genuinely active and genetically implicated in Sjögren's-associated lymphoma (P 26338037 P 24159176), P 30067782 but it is confined to a biomarker-identifiable minority (P 29669392), one pillar of the stated premise is factually wrong for pSS (P 35205758 P 19961773), no MALT1 inhibitor has ever been tested in the disease, the closest upstream agents move only objective — not symptom — endpoints (P 37932009 P 38073013),

and the drug's own mechanism carries a Treg-depleting, autoimmunity-provoking liability (P 32425939) that is acutely problematic in an autoimmune indication. The single most efficient test is **not** a clinical trial but a preceding **ex vivo biopsy-explant study** in enriched, GC-positive/cryoglobulinemic patients that measures, side by side, suppression of pathogenic B-cell NF-κB signalling versus impairment of Treg function under safimaltib — the experiment that most cheaply resolves whether on-target benefit can be separated from on-target harm. Only if that favours the drug should a small, biomarker-enriched, objective-endpoint Phase 1b/2a proceed.

PMIDs / identifiers carrying the most weight:

- PMID: 26338037 — Germline TNFAIP3 variant links impaired NF-κB control to pSS lymphoma (OR 2.48); strongest human-genetics support.
- PMID: 24159176 — A20 functional abnormalities in 77% of pSS-MALT lymphoma; the target population exists.
- PMID: 30067782 — Epithelial A20 loss reproduces Sjögren-like gland disease; causal pathway proof in vivo.
- PMID: 32425939 — MALT1 inhibition depletes Tregs and causes IPEX-like autoimmunity; the decisive safety counter-argument.
- PMID: 35205758 — pSS-MALT lymphomas lack the classic MALT1 aberrations; refutes a stated premise.
- PMID: 37932009 — Remibrutinib (NCT04035668): upstream pathway blockade improves ESSDAI but not symptoms; best clinical proof-of-concept and endpoint warning.
- PMID: 42814116 — Safimaltib Phase 1 (NCT03900598): modest ORR 28.8%, BCR-mutant responders; defines efficacy ceiling and enrichment strategy.
- PMID: 29669392 — Ectopic GCs confer 7.8× lymphoma risk; the enrichment biomarker any trial must use.
- PMID: 24004675 — MALT1 inhibitors validated in ABC-DLBCL; autoimmune use explicitly only a "perhaps."

Established findings are drawn from the cited primary literature and trial records. Interpretive statements synthesizing those findings into a repurposing verdict are labelled as inference and represent this assessment's reasoning, not direct experimental results.